Piyush Majee

+91-9625103514 piyushmajee28@gmail.com LinkedIn Github

Education

Indian Institute of Technology Delhi Delhi, India
Bachelor of Technology in Engineering Physics *2021 – 2025*

- **Relevant Coursework:** Data Structures & Algorithms, Probability & Stochastic Processes, Linear Algebra & ODE, Signals & Systems, Digital Electronics, Quantum Information & Computing, Computational Physics

Experience

Member of Technical Staff - I | Oracle July 2025 – April 2026

Oracle Service Bus

- Delivered an **Oracle ADF**-based console for authorization **policy management**, reducing configuration time by **70%**
- Engineered a **policy expression engine** with **XML processing** and **JMX** integration to persist **WebLogic** policies
- Shipped **multipart/form-data** support for **REST APIs** using **Java** and **Jersey**, eliminating binary attachment loss
- Eliminated **2 GB payload limit** in **REST services** via **streaming I/O**, enabling reliable transfer of large payloads
- Resolved **50+ production issues** across API integrations, message processing, and distributed OSB/SOA environments through **root cause analysis**, improving product reliability

TriageX

- Designed **microservice-based** bug triage platform on **OCI** using **LangChain**, reducing bug analysis time by **40%**
- Engineered a **REST** ingestion service to preprocess, chunk, and generate **vector embeddings** for **5k+** bug documents
- Boosted **retrieval hit-rate to ∼80%** by implementing **hybrid search (dense + BM25)** with **metadata filtering**
- Deployed full-stack UI for querying the **RAG** pipeline with dynamic **Top-K** tuning, adopted by **4+ internal teams**

Software Developer Intern | Oracle May 2024 – July 2024

- Built a **test analytics dashboard** to visualize **integration test** execution results, reducing debugging time by **∼30%**
- Migrated the **integration test framework** and resolved **25+ failing test cases**, restoring **end-to-end test coverage**
- Created a **Jenkins CI/CD pipeline** and automated **test environment provisioning**, reducing setup time by **∼85%**

Projects

Code Review Agent May 2026

- Built a **multi-agent** PR reviewer using **LangGraph** with parallel specialist agents and confidence-gated result aggregation
- Engineered an event-driven review pipeline using **FastAPI**, **Redis**, and **ARQ** with secure **GitHub webhook** ingestion
- Designed a **hybrid RAG** memory layer using **PostgreSQL**, **pgvector**, and **TimescaleDB** for scalable agent execution

OrderFlow March 2026

- Architected an **event-driven** microservices platform with **Spring Boot**, **Kafka**, and **PostgreSQL** using the **Saga pattern**
- Developed a responsive **Next.js** frontend with **JWT authentication** and real-time order tracking using **WebSockets**
- Containerized microservices with **Docker** and deployed them on **AWS ECS Fargate** using **Amazon ECR**

FraudGuard April 2026

- Built an end-to-end credit card fraud detection service using **Python**, **FastAPI**, and **Docker** for real-time evaluation
- Trained **XGBoost** on 284k transactions with SMOTE & custom thresholds to handle 0.17% imbalance at **0.88 PR-AUC**
- Optimized the inference pipeline using **ONNX Runtime**, reducing model latency to **sub-10ms** on CPU

Shell Design 2024

- Engineered a **Unix-like shell** in **C++** using **POSIX** APIs for process management, **IPC**, piping, **I/O redirection**
- Developed Linux utilities using **procfs** for process inspection, **lock-aware file deletion**, and **process tree traversal**
- Implemented **heuristic-based malware detection** using process hierarchy, CPU utilization, and runtime metrics

Technical Skills

Languages: C++, Python, Java, SQL, JavaScript | **Concepts**: Microservices, Database, Concurrency, System Design
Backend: Spring Boot, JUnit, FastAPI, PostgreSQL, MongoDB, Kafka, AWS/OCI, Jenkins, Linux, Docker, Kubernetes
AI/ML: PyTorch, NumPy, Pandas, Deep Learning, NLP, LLMs, Model Deployment, LangChain, RAG, MCP, Transformers